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DemandIQ

Demand Planning, IBP & New Product Launch Decision System for Premium Outdoor Apparel

From stockout-adjusted forecasting and weekly S&OE decisions to cold-start launch planning, rolling forecasts, and lifecycle handoff.

DEMAND PLANNING | FORECASTING | IBP | S&OE | NEW PRODUCT PLANNING

Portfolio simulation built using public, synthetic, and derived data. No real company internal planning data was used.

DemandIQ at a Glance

Everything you need to read the rest of this document.

A | PRODUCTS

MATURE | OWN HISTORY

APS-001

Alpine Performance Shell

CTS-001

Core Technical Shell

IMH-001

Insulated Midlayer Hoody

NEW | NO OWN HISTORY

HIS-001

Hybrid Insulated Shell

HIS-001 has no demand history of its own at launch. It follows a cold-start, analog-led process instead of the mature forecasting engine.

B | CHANNELS

ECOM

Direct online

RETAIL

Own stores

WHOLESALE

Partner

D | WHAT IS DEMANDIQ?

A synthetic premium outdoor apparel demand-planning simulation.

It shows how a planner moves from retail demand signals to decisions: demand reconstruction, forecasting, scenario planning, inventory and service risk, and governed planner actions. It then extends the same environment to a new-product launch with analog selection, an 18-month cold-start forecast, IBP consensus, the initial buy, sell-through, reforecasting, FVA, and rolling updates. Every figure comes from public, synthetic, or derived data. Economics are planning exposure proxies.

C | ESSENTIAL PLANNING TERMS

WAPE

Weighted Absolute Percentage Error. Lower is better.

Bias

Whether forecasts run above or below demand.

IBP

Integrated Business Planning, the consensus plan.

S&OE

Sales and Operations Execution, weekly execution.

FVA

Forecast Value Add. Did a step improve accuracy?

V0 to V3

Analytical, Commercial, Consensus, Approved.

WOS

Weeks of Supply. How long inventory lasts.

Fill Rate

Share of demand inventory can serve.

Cold Start

Forecasting a product with little or no history.

Why Planning Is More Than Predicting Sales

THE PLANNING CHALLENGE

- Sales can be censored by stockouts, so observed sales can understate real demand.
- Weather adds uncertainty, but future weather is not known at forecast time.
- Aggregate fill can hide weekly service failures.
- A forecast has to become an inventory decision.
- New products have no history of their own.
- The plan has to learn as evidence arrives through governed reforecasting.

WHAT I FOUND

MATURE 13-WK BASE DEMAND 36,435 units portfolio	PORTFOLIO BASE FILL 98.77% vs 92% target
WEEKLY-SERVICE P1 SERIES 3 of 9 escalate	HIS-001 V3 APPROVED 13-WK ~7,577 unconstrained demand
HIS-001 BUY LAUNCH FILL ~8,259 units ~97.7% fill	CYCLE-02 LIKE-FOR-LIKE -3.05% rolling revision

The point of the project

Aggregate accuracy and aggregate fill are necessary but not sufficient. The value sits in the decision: which weeks and channels are at risk, what to buy for a product with no history, and how the plan should learn after launch.

MATURE ENGINE

NEW PRODUCT ENGINE

PLANNER GOVERNANCE

How DemandIQ Fits Together

Two governed engines share one planning environment.



Recover the Demand Signal, Then Forecast It Honestly

A | WHY SALES ARE NOT ALWAYS DEMAND

When inventory runs short, sales are censored. If I forecast those sales directly, the model learns that stockout weeks were low-demand weeks and plans down the products that sold best. So I reconstructed demand before forecasting and chose Seasonal Profile Imputation under one governed evaluation.

The forecast target is reconstructed demand. A hidden true-demand series only scores the reconstruction. It is never a forecast input.

CENSORED-ROW WAPE

26.08%

BIAS

-2.61%

LOST-DEMAND RECOVERY

95.68%

FULL-STOCKOUT RECOVERY

98.43%

B | THE MATURE FORECAST ENGINE

Seasonality	52 weeks
Initial training	104 weeks
Forecast horizon	13 weeks
Expanding folds	12
Primary metric	WAPE bias tracked

CHAMPION MIX

8 ETS

+ 1 baseline
0 SARIMA

CHAMPION WAPE

8.4 to 21.7%

bias -3.1% to +2.4%

I chose the defensible model, not the most complex one

Where SARIMA beat ETS by about 0.06 points of WAPE, I kept the simpler ETS. Where a moving-average baseline won outright, I kept it. Repeated out-of-sample performance, bias, and simplicity drove the choice.

From 13-Week Outlook to Planner Decisions

A | WEATHER AND DEMAND OUTLOOK

MILD

36,036

units

BASE

36,435

units

SEVERE

36,677

units

Weeks 1-3 need a real nowcast. None was supplied, so I made no near-term adjustment. Weeks 4-13 use Mild, Normal, and Severe scenarios. No realized future weather is used.

B | SERVICE AND INVENTORY RISK

PORTFOLIO BASE FILL

98.77%

looks healthy

SERVICE TARGET

92%

weekly target

P1 SERIES

3 of 9

weekly-service

COVERAGE

All 9

below 2.5-wk policy

Aggregate health hides weekly risk

98.77% portfolio fill looks solved. Yet three of nine series miss the weekly target twice. All three share a worst week of 2026-08-24, when committed receipts are zero across all nine series. I treat the timing as a signal, not proven cause.

P1 SERIES	13W fill	Worst week	Action
APS-001 WHOLESALE	93.5%	33.2%	ESCALATE
CTS-001 RETAIL	95.6%	65.4%	ESCALATE
IMH-001 WHOLESALE	96.1%	74.1%	ESCALATE

Risk detection is not execution authorization

Three series escalate and six protect. The engine releases zero automatic chase and zero reallocation because supplier lead time, expedite, and transfer feasibility are not modeled. Base lost-revenue exposure is about CAD 192K, a planning exposure proxy used only to rank the review queue.

HIS-001: Forecasting a Product With No History

NEW PRODUCT

Hybrid Insulated Shell | launched 31 Aug 2026

It had no demand history of its own, so the mature engine could not be applied.

A | ANALOG SELECTION

APS-001		0.679	Primary 60%
CTS-001		0.531	Excluded
IMH-001		0.448	Secondary 40%

Why 60% APS + 40% IMH

APS anchors shell construction, price, scale, and Fall/Winter seasonality. IMH supplies the insulation that APS lacks, and the two share a similar seasonal shape. CTS ranked second by raw score but is a redundant shell that fills no gap, so I left it out.

B | THE V0 COLD-START FORECAST

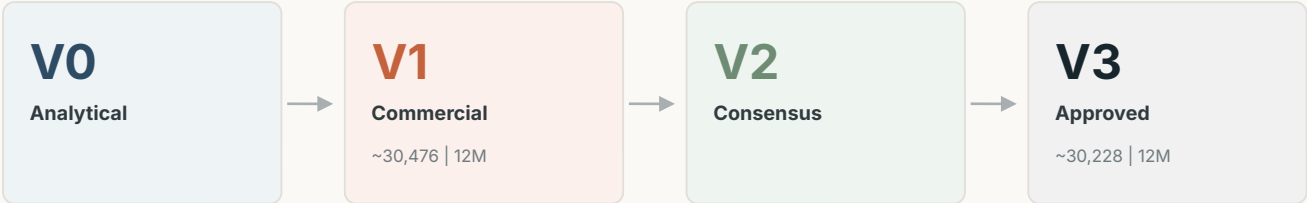
13-WEEK V0 ~6,888 units	FIRST-YEAR V0 ~27,480 units	18-MONTH V0 ~44,919 units	SECOND-SEASON FACTOR 1.00 no growth before evidence
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C | WHY TWO HORIZONS

STRATEGIC IBP 18 months, monthly Commercial, category, and supply discussions.	OPERATIONAL S&OE 13 weeks, weekly Near-term launch execution and the initial buy.
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From Consensus to an Approved Plan and a Buy

A | THE FORECAST VERSION JOURNEY



V3 is approved, unconstrained demand

V3 is about 1.10 times V0. First 12 months are about 30,228 units and 18 months about 49,411. It is a demand position, not a supply-constrained plan.

B | THE INITIAL BUY

Approved V3 13-week demand	~7,577
+ Launch uncertainty buffer (9%)	~682
Frozen initial buy BALANCED	~8,259
Flex reserve held back	~991

C | CHANNEL ALLOCATION

ECOM		~3,271
RETAIL		~2,544
WHOLESALE		~1,454
FLEX RESERVE		~991

Launch buffer is not mature safety stock

The 9% buffer is a one-time cold-start allowance for launch uncertainty. It is not the mature 2.5-week safety-stock rule. Synthetic supply planning assumptions use about an 8-week replenishment lead and chase capacity of 15% of the buy or less.

What Happened After Launch

SYNTHETIC SEEDED LAUNCH SIMULATION | SEED 7

OBSERVED SALES

~6,650

13 weeks

LATENT DEMAND | EVAL ONLY

~6,809

hidden truth

LAUNCH FILL

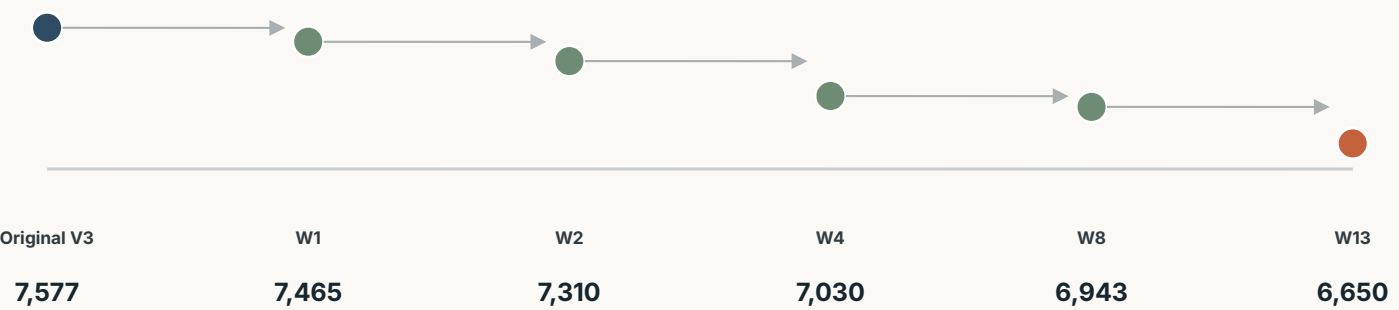
~97.7%

observed sell-through

Evaluation boundary

The latent demand is for evaluation only. The operational planner never saw hidden demand truth, and it never entered a reforecast or decision.

B | THE REFORECAST JOURNEY | PLANNER LEARNING OVER TIME



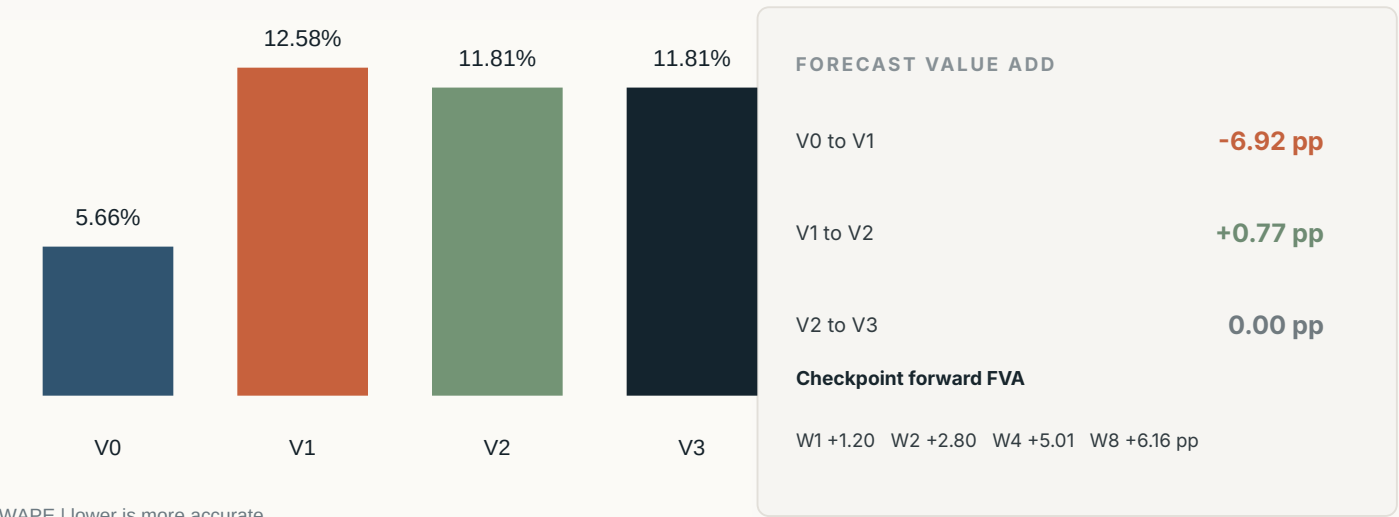
As launch evidence built up, the plan leaned less on the pre-launch analog prior and more on what was actually selling. The weighting rises with observed weeks ($w = n / (n + 4)$). The planning point matters more than the formula: real evidence earns weight over time.

A disciplined non-action

Every checkpoint resolved to hold. Attainment stayed within tolerance, channel-mix deviation stayed under the reallocation trigger, and coverage stayed adequate. I forced no chase, cut, or reallocation. That is disciplined governance, not a missing feature.

Did the Forecast Add Value, and What the Channels Taught Me

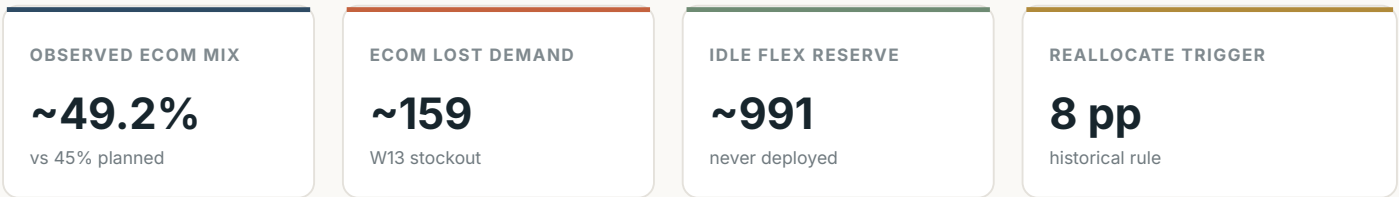
A | DID EACH VERSION IMPROVE ACCURACY?



A caveat I will not hide

The synthetic launch generator was centered on the V0 analytical baseline. So V0 scoring better than V1, V2, and V3 on this seeded path is an illustrative governance demonstration. It is not independent proof that commercial input was harmful.

B | WHAT THE CHANNELS TAUGHT ME | OBSERVED



Two different problems

Inventory sat available in the reserve and other channels while ECOM stocked out. That is a channel-allocation and policy-timing issue, which I recorded as a next-cycle learning. I do not claim another threshold is optimal.

Rolling the Forecast, and Where HIS-001 Sits

A | A GENUINE ROLLING FORECAST



What changed

September becomes actual. March 2028 enters as a newly generated month 18, built from the same governed seasonal foundation rather than copied. Across the 17 overlapping months, the outlook falls by about 3.05%.

LIKE-FOR-LIKE REVISION

-3.05%

17 overlap months

LIFECYCLE LEARNING WEIGHTS

Near-term 1.00 Normalization 0.50 Second season 0.25

Synthetic rolling-forecast governance assumptions

B | LIFECYCLE HANDOFF



CURRENT STATUS

EARLY_LAUNCH

13 observed weeks

MATURE-MODEL ELIGIBLE?

NO

no ETS/SARIMA yet

Why new is not mature

Thirteen weeks is not mature history. The current method stays an analog plus actual-evidence blend because new products and mature products need different governance.

What This Project Shows

CAPABILITIES

FORECASTING

Reconstruction
Backtesting
Accuracy & bias
Weather scenarios

PLANNING & S&OE

13-week S&OE
Inventory / service risk
Exception management
Planner governance

IBP / NEW PRODUCT

Consensus & versions
Analog / cold start
Initial buy / allocation
Rolling forecast / handoff

GOVERNANCE

- Public, synthetic, and derived data. No real company internal data.
- Hidden truth is used for evaluation only.
- Economics are planning exposure proxies, not profit.
- Enterprise concepts are demonstrated with no unsupported platform-experience claim.

LIMITATIONS

- A synthetic launch on a single seeded path.
- Simplified supplier constraints; weather caps are assumptions.
- FVA shaped by a simulation centered on V0.
- 13 launch weeks, too few to fit a mature model.

The takeaway

DemandIQ moved the question from "What is likely to sell?" to something a planner can act on:

"What demand should we plan for, what inventory and service risk does that create, what action should the planner take, how should a new product be launched, and how should the plan learn as new evidence arrives?"

The value is in planning decision quality and governance, not predictive accuracy alone.